AI Assisted Coding ASSIGNMENT-9.5

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Batch: 45

Problem 1: String Utilities Function

Consider the following Python function:

def reverse_string(text):

return text[::-1]

Task:

1. Write documentation in:

o (a) Docstring

o (b) Inline comments

o (c) Google-style documentation

2. Compare the three documentation styles.

3. Recommend the most suitable style for a utility-based string

library.

PROMPT:

Generate documentation for the following Python function in three styles:

1) Standard docstring

2) Inline comments

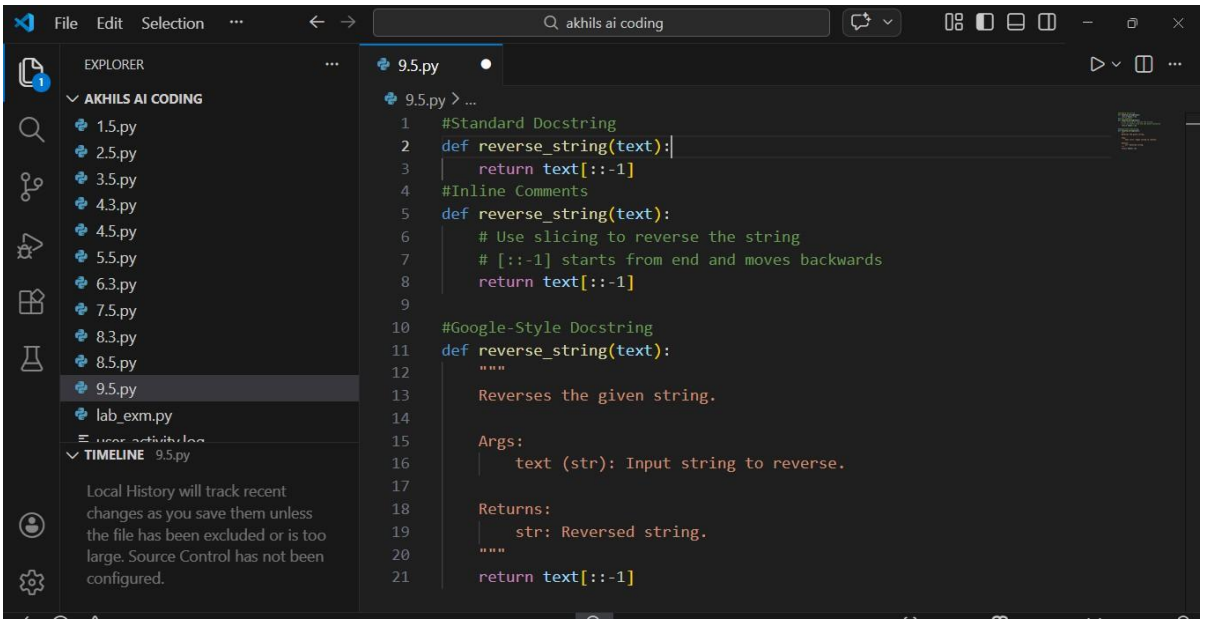
3) Google-style docstring

Function:

```
def reverse_string(text):
    return text[::-1]
```

Also compare the three styles and recommend the best style for a utility-based string library.

OUTPUT:



Problem 2: Password Strength Checker

Consider the function:

def check_strength(password):

return len(password) >= 8

Task:

1. Document the function using docstring, inline comments, and

Google style.

2. Compare documentation styles for security-related code.

3. Recommend the most appropriate style.

PROMPT:

Document the following security-related function using:

1) Standard docstring

2) Inline comments

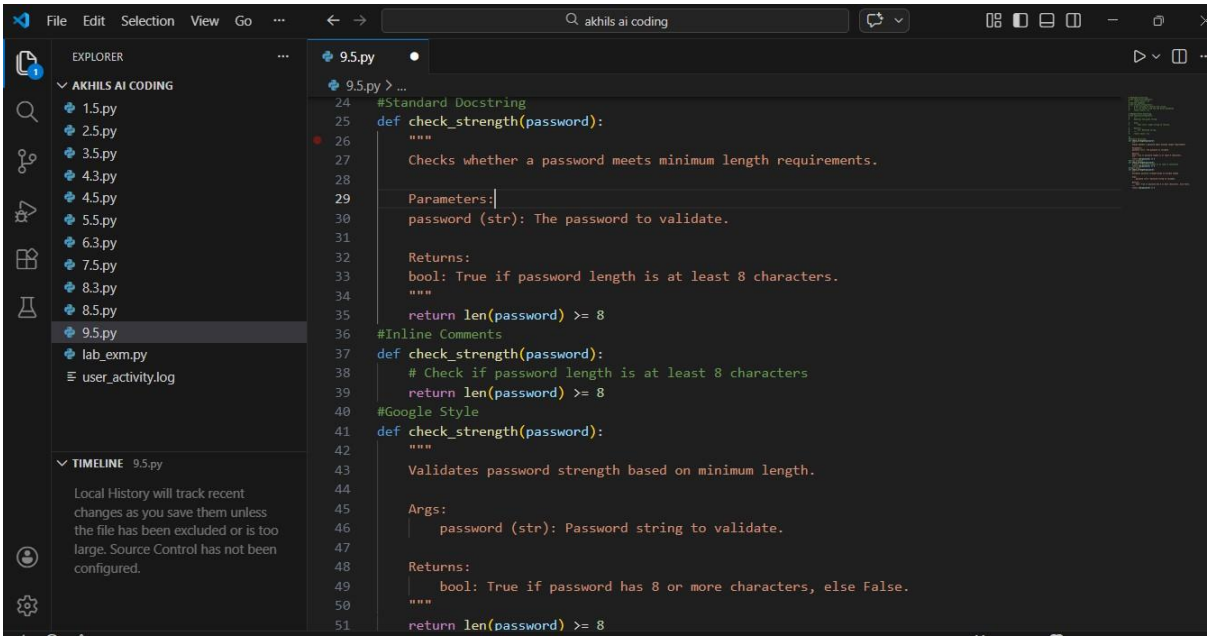
3) Google-style documentation

Function:

```
def check_strength(password):
    return len(password) >= 8
```

Compare documentation styles for security-related code and recommend the best one.

OUTPUT:



Problem 3: Math Utilities Module

Task:

1. Create a module math_utils.py with functions:

o square(n)

o cube(n)

o factorial(n)

2. Generate docstrings automatically using AI tools.

3. Export documentation as an HTML file.

PROMPT:

Create a Python module named math_utils.py with functions:

- square(n)

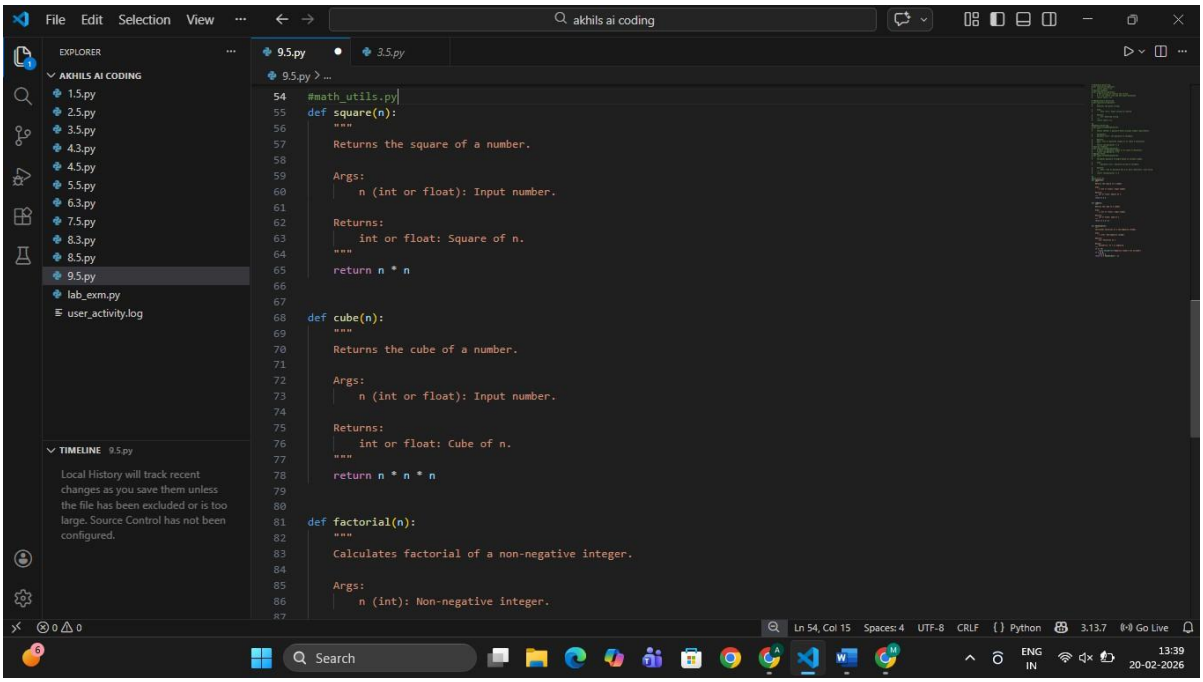
- cube(n)

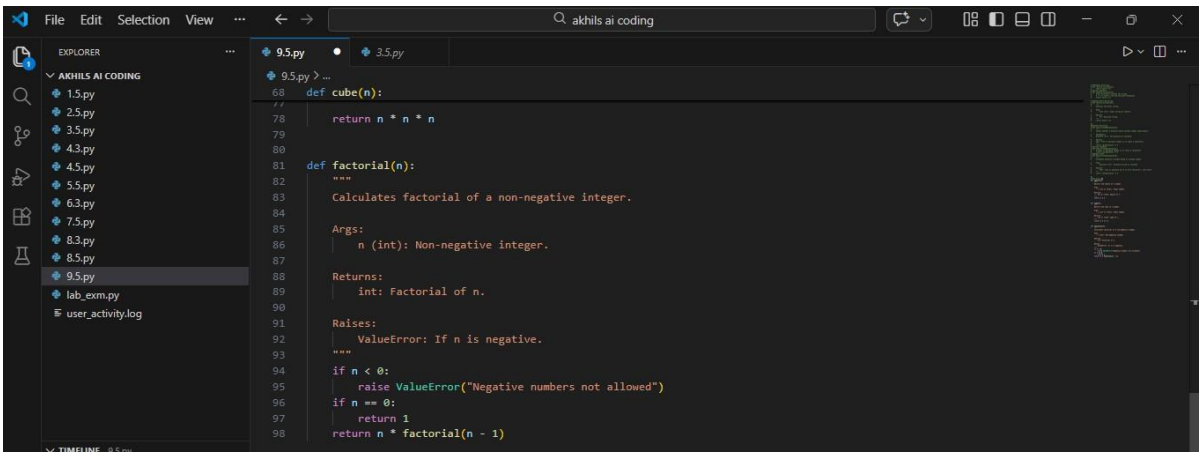
- factorial(n)

Add Google-style docstrings.

Generate documentation using Sphinx and export it as HTML.

OUTPUT:





Problem 4: Attendance Management Module

Task:

1. Create a module attendance.py with functions:

o mark_present(student)

o mark_absent(student)

o get_attendance(student)

2. Add proper docstrings.

3. Generate and view documentation in terminal and browse

PROMPT:

Create attendance.py module with:

- mark_present(student)

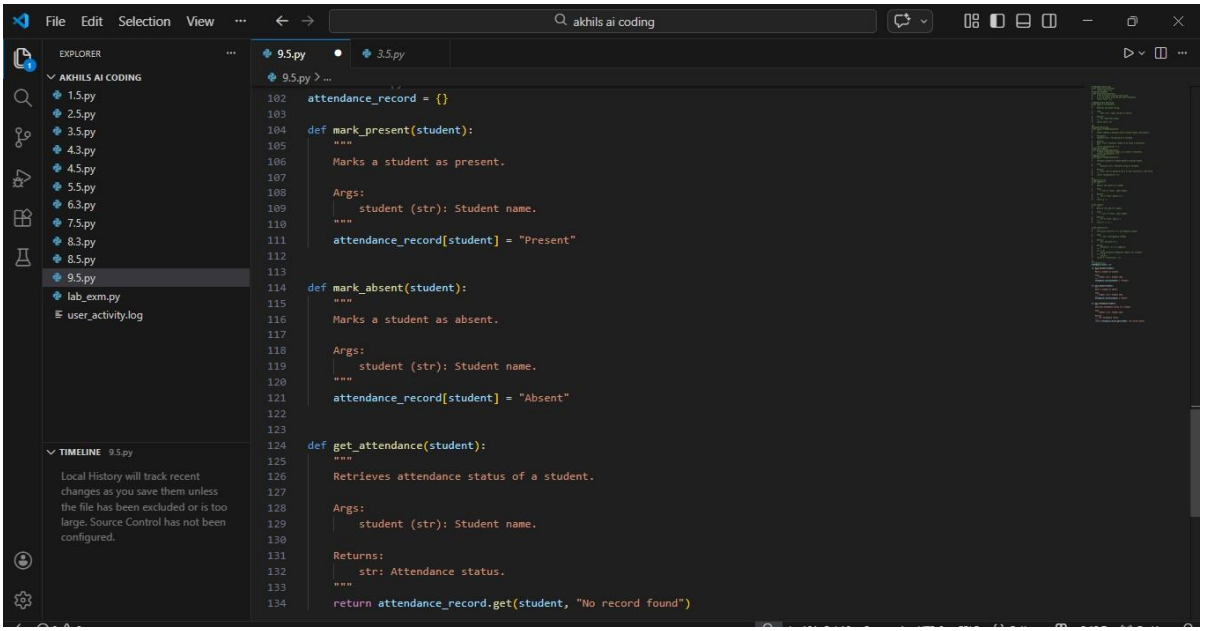
- mark_absent(student)

- get_attendance(student)

Add Google-style docstrings.

Generate documentation and show how to view in terminal and browser.

OUTPUT:



Problem 5: File Handling Function

Consider the function:

def read_file(filename):

with open(filename, 'r') as f:

return f.read()

Task:

1. Write documentation using all three formats.

2. Identify which style best explains exception handling.

3. Justify your recommendation.

PROMPT:

Document the following function in:

1) Standard docstring

2) Inline comments

3) Google-style documentation

Function:

```
def read_file(filename):
    with open(filename, 'r') as f:
        return f.read()
```

Identify which style best explains exception handling and justify.

OUTPUT: